

Spectroscopic study of low-lying ^{16}N levels

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The magnitude of the $^{15}\text{N}(n, \gamma)^{16}\text{N}$ reaction rate in asymptotic giant branch stars depends directly on the neutron spectroscopic factors of low-lying ^{16}N levels. A new study of the $^{15}\text{N}(d, p)^{16}\text{N}$ reaction is reported populating the ground and first three excited states in ^{16}N . The measured spectroscopic factors are near unity as expected from shell model calculations, resolving a long-standing discrepancy with earlier measurements that had never been confirmed or understood. Updated $^{15}\text{N}(n, \gamma)^{16}\text{N}$ reaction rates are presented.

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The astrophysical production site of galactic ^{19}F is rather uncertain with asymptotic giant branch (AGB) stars, supernovae, and Wolf-Rayet stars as suggested possibilities [1]. AGB stars seem the most likely as they are the only astrophysical site observationally confirmed to produce ^{19}F [2]. In fact, recent observations [3] have found F overabundances of up to 250 times solar values in extremely hot post-AGB stars. Model calculations, however, have not yet been able to reproduce many observed abundance correlations such as that of ^{19}F with ^{12}C [2], and it is unclear the extent to which nuclear physics uncertainties are contributing to the discrepancies.

Galactic ^{19}F is rather rare, in part, because it is easily destroyed by p - and α -induced reactions in the astrophysical environment. A detailed understanding of the nuclear reactions occurring in the stellar site is therefore required to model ^{19}F production. ^{19}F is thought to be produced in AGB stars via the sequence $^{14}\text{N}(\alpha, \gamma)^{18}\text{F}(\beta^+)^{18}\text{O}(p, \alpha)^{15}\text{N}(\alpha, \gamma)^{19}\text{F}$. Free protons and neutrons are liberated by the $^{14}\text{N}(n, p)^{14}\text{C}$ and $^{13}\text{C}(\alpha, n)^{16}\text{O}$ reactions, respectively, and further processing by p - and n -induced reactions must also be considered [2,4]. In particular, one needs to consider the branching at ^{15}N between the $^{15}\text{N}(\alpha, \gamma)^{19}\text{F}$ and $^{15}\text{N}(n, \gamma)^{16}\text{N}$ reactions when calculating ^{19}F production. While there has been considerable recent work studying the $^{15}\text{N}(\alpha, \gamma)^{19}\text{F}$ reaction [1,5,6], there have been few studies of the $^{15}\text{N}(n, \gamma)^{16}\text{N}$ reaction. The most recent evaluation of the $^{15}\text{N}(n, \gamma)^{16}\text{N}$ reaction rate was performed in 1996 by Meissner *et al.* [7], where it was found that the rate is dominated by p -wave direct capture over almost all of the relevant temperature range with resonances only contributing at the highest of temperatures. The magnitude of the calculated direct capture rate depends directly on the neutron spectroscopic factors of low-lying ^{16}N levels, and thus a determination of these spectroscopic factors is critical for calculating the rate.

There has been a single measurement of the neutron-spectroscopic factors for low-lying ^{16}N levels [8]. In that

study, 5–6 MeV deuterium beams bombarded enriched-in- ^{15}N melamine ($\text{C}_3\text{H}_6\text{N}_6$) targets, and (d, p) reaction protons were detected in silicon detectors. The ground state of ^{16}N and excited states at 121, 297, and 397 keV were observed with nearly equal spectroscopic factors of ~ 0.5 . This result was somewhat surprising as ^{15}N is a closed shell nucleus, and it was thought that low-lying ^{16}N levels were good single-particle levels with spectroscopic factors near unity [7,8]. These expectations were further confirmed by OXBASH calculations in Ref. [7] where spectroscopic factors between 0.87 and 0.96 were predicted. Additional information comes from a study of low-lying levels in the mirror nucleus, ^{16}F , where it was found that the observed properties of the mirror levels were only consistent with proton spectroscopic factors near unity [9]. If isospin symmetry holds, this would again imply that the neutron spectroscopic factors in ^{16}N should be near unity. Clearly, the low spectroscopic factors for this nucleus need experimental confirmation.

To investigate these discrepancies, a new study of the $^{15}\text{N}(d, p)^{16}\text{N}$ reaction has been performed at the Oak Ridge National Laboratory (ORNL) Holifield Radioactive Ion Beam Facility (HRIBF). An inverse kinematics (a heavy beam on a light target) study was chosen because (1) a well-developed infrastructure has been developed to study (d, p) reactions in inverse kinematics [10–12], (2) enriched ^{15}N targets can be difficult to fabricate while CD_2 targets are readily available and well characterized [13,14], and (3) the use of inverse kinematics readily allows measurements at the most forward center-of-mass angles where the cross section is most sensitive to the spectroscopic factors, while normal kinematics (d, p) measurements can be difficult near 0° where the rate of elastically-scattered deuterons is high.

A 100-MeV ^{15}N beam was used to bombard a $90\text{-}\mu\text{g}/\text{cm}^2$ CD_2 target. Similar to previous studies [10], protons from the (d, p) reaction were detected at backward laboratory angles, $155^\circ\text{--}169^\circ$ (corresponding to $3^\circ\text{--}8^\circ$ in the center of mass)

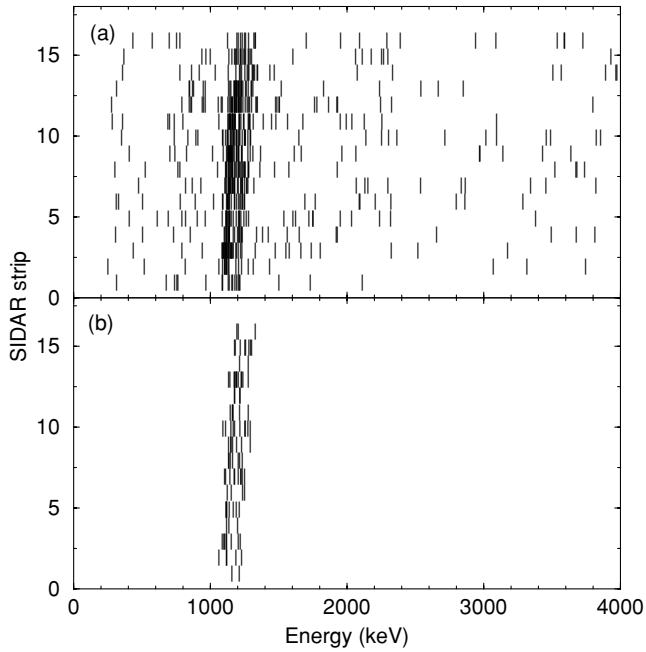


FIG. 1. (a) Energy of detected particles observed in the various SIDAR strips. The laboratory angles range from 169° to 155° for strips 1 to 16, respectively. (b) Same as (a) but in coincidence with a ^{16}N recoil transported through the DRS. The band arising from $^{15}\text{N}(d, p)^{16}\text{N}$ is clearly identified.

by the Silicon Detector Array (SIDAR) [15]. To ensure that the identification of (d, p) protons was correct, some of the data were taken in coincidence with ^{16}N recoils that were transported and separated from the primary beam by the Daresbury Recoil Separator [16]. Figure 1 shows the spectrum from SIDAR in singles and in coincidence with ^{16}N recoils. Additional detectors of the Oak Ridge Rutgers University Barrel Array (ORRUBA) type [17] were placed near 90° and were useful for detecting (d, p) protons at larger center of mass angles and for monitoring target stability via elastic scattering of target components during the run (Fig. 2). The ORRUBA strips were oriented parallel to the beam axis such that the position along the strip of a detected particle provided a good measure of the polar angle of the reaction product. Another smaller annular detector was placed at forward angles (9° – 18°) to measure elastic scattering of the beam from the carbon in the target for beam current normalization. To determine this scattering rate, a silicon detector was inserted into the beam path downstream of the target location, and a low-intensity beam ($\sim 10^4$ pps) was counted while scattered beam particles were measured in the forward-angle detector. Afterwards, the beam detector was retracted, and the beam was raised to full intensity. Typical beam intensities were kept below 3×10^6 $^{15}\text{N}/\text{s}$ to prevent target degradation. The lack of target degradation was verified by monitoring the rate of deuterium scattering (Fig. 2) over time and by checking that consistent target thickness measurements were obtained before and after the experiment.

Data were taken for approximately 65 h and a total of 4.6×10^{11} ^{15}N ions impinged on the target over the course of the experiment. A typical singles spectrum from one of the

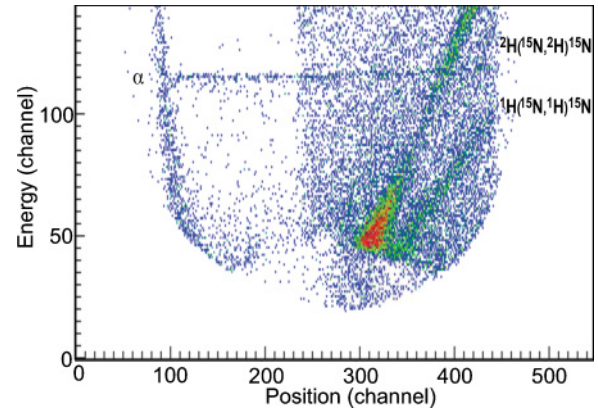


FIG. 2. (Color online) Detected energy vs. position spectrum for a portion of the events observed in an ORRUBA detector placed near 90° . The scattering of target constituents is noted at positions corresponding to laboratory angles forward of 90° . These scattering rates provide a sensitive measure of target composition changes—none of which were observed. The line from the α -calibration source is clearly visible since the energy of these events show no position dependence. The reduction of counts backwards of 90° arises from target-frame shadowing.

SIDAR strips is shown in Fig. 3. Two groups were observed corresponding to the ground state and first excited state in one group and the second and third excited states in the other group. While the resolution was not quite sufficient to separate the states, this did not negatively impact the goal of determining the total cross sections.

Cross sections were extracted for the population of these low-lying ^{16}N levels. Because coincidence data with ^{16}N recoils were only taken for part of the experiment, the singles spectra were used for peak sum extraction assuming a linear background. As can be seen in Fig. 3, there was relatively little background in the spectra, and thus background subtraction was straightforward. The solid angle coverage of the SIDAR array was determined using a calibrated α source placed

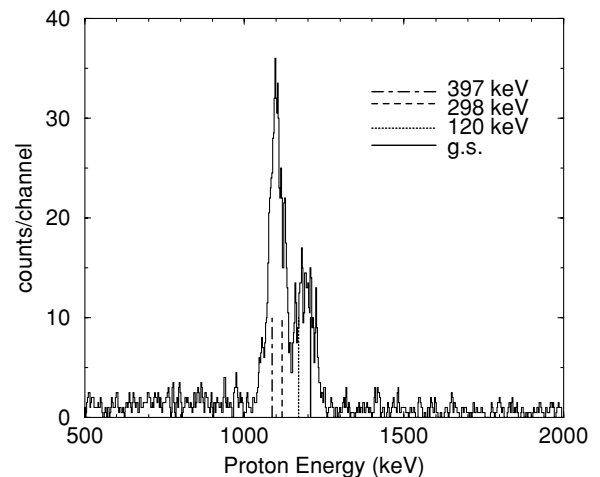


FIG. 3. Singles energy spectrum observed in the inner strip of SIDAR. Lines show the expected positions of peaks from the $^2\text{H}(^{15}\text{N}, p)^{16}\text{N}$ reaction.

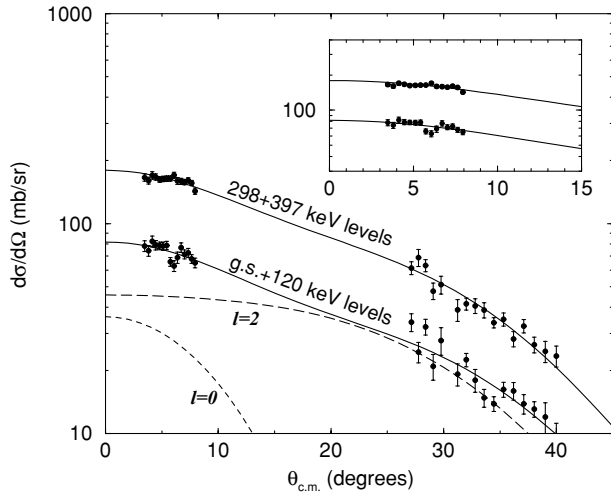


FIG. 4. Differential cross sections for the $^{15}\text{N}(d, p)^{16}\text{N}$ reaction as a function of center-of-mass angle. The inset shows an expanded view of the SIDAR data. Since resolution of the closely-spaced levels was not possible, composite DWBA curves have been fitted to the data. The dashed curves show the contributions that were summed to fit the g.s.+120-keV cross sections.

at the target position. The solid angle calibration using the source agreed with geometrical calculations within 1–4% typically. The target thickness was determined from energy loss measurements of 5.8-MeV α particles and from the weight of deuterated polyethylene that was originally dissolved in p-xylene to produce the target. The two methods agreed within 11%. The combined differential cross sections for populating the ground and first excited states and for populating the second and third excited states are shown separately in Fig. 4. The uncertainties shown on the data points are purely statistical in nature. The cross sections extracted at the most forward center of mass angles were from the SIDAR detector, while those at larger angles were measured with the ORRUBA detectors.

The differential cross sections determined from the present measurements were analyzed within the framework of the distorted wave Born approximation (DWBA) using global optical model parameter sets. The deuteron and proton parameters of Perey and Perey [18] were found to be well suited for the data, and these same parameters also reproduced well the published DWBA calculations from Ref. [8]. Table I summarizes the spectroscopic properties of the low-lying

TABLE I. Spectroscopic properties of ^{16}N excitations in comparison with shell model predictions [7]. The last two columns show the results of the present work when the magnitudes of the individual components were fixed to the ratios obtained in Ref. [8] or were allowed to vary freely, respectively. Statistical uncertainties are given in parenthesis and the systematic uncertainties are estimated to be $\sim 15\%$.

$E_x(\text{MeV})$	J^π	nl_j	C^2S Ref. [8]	C^2S OXBASH [7]	C^2S present ^a	C^2S present ^b
0	2^-	$1d_{5/2}$	0.55	0.93	0.96(2)	1.04(4)
0.120	0^-	$2s_{1/2}$	0.46	0.95	0.80(1)	0.71(6)
0.298	3^-	$1d_{5/2}$	0.54	0.87	0.91(1)	1.03(3)
0.397	1^-	$2s_{1/2}$	0.52	0.96	0.88(1)	0.74(5)

^aRatio of components constrained in fit.

^bComponents allowed to vary freely.

states in ^{16}N and compares the present extracted spectroscopic factors (C^2S) with previous measurements [8] and OXBASH shell model calculations [7]. Table II summarizes the optical model parameters used in the DWBA calculations. Calculations of the cross sections to populate individual ^{16}N levels were performed with the reaction code TWOFNR [19]. Since closely-spaced levels could not be resolved, the calculations were combined and fitted to the data using two different methods. First, the individual components were weighted by the relative spectroscopic factors reported in Ref. [8] with the overall magnitude allowed to vary in the fit. The resulting spectroscopic factors are shown in column 6 of Table I with the statistical uncertainty from the fit shown in parenthesis. Second, the magnitudes of the individual components were allowed to vary freely. The resulting spectroscopic factors from this method are shown in the last column of Table I and the best fit is shown in Fig. 4. Both methods produce reasonable fits to the data and spectroscopic factors that agree much better with the near-unity OXBASH calculations [7]. The systematic uncertainties in the spectroscopic factors are estimated to be about 15% resulting mostly from uncertainties in the target thickness ($\sim 11\%$) and beam current normalization ($\sim 10\%$). Additional model uncertainties are not considered. We do not believe that the difference in our spectroscopic factors and those extracted by [8] are the result of using different optical model parameters as we reproduce the DWBA calculations in Ref. [8] using our parameters.

The $^{15}\text{N}(n, \gamma)^{16}\text{N}$ reaction rate calculation of Ref. [7] has been updated using the average of the spectroscopic

TABLE II. Global optical model parameters of Ref. [18] used as input into the DWBA code TWOFNR [19].

	V^a (MeV)	r_0 (fm)	a_0 (fm)	W (MeV)	W_D (MeV)	r_W (fm)	a_W (fm)	V_{so} (MeV)	r_{so} (fm)	a_{so} (fm)	r_C (fm)
$^{15}\text{N}+d$	85.31	1.15	0.81	0.0	16.0	1.34	0.68	0.0	1.0	1.0	1.15
$^{16}\text{N}+p$ (g.s.)	54.19	1.25	0.65	0.0	13.5	1.25	0.47	7.5	1.25	0.47	1.25
n	^b	1.25	0.65	0.0	0.0	—	—	6.0	1.25	0.65	1.25

^aThe parameter definitions follow the normal conventions and correspond to those found in Ref. [18].

^bFit to reproduce the binding energy of the neutron.

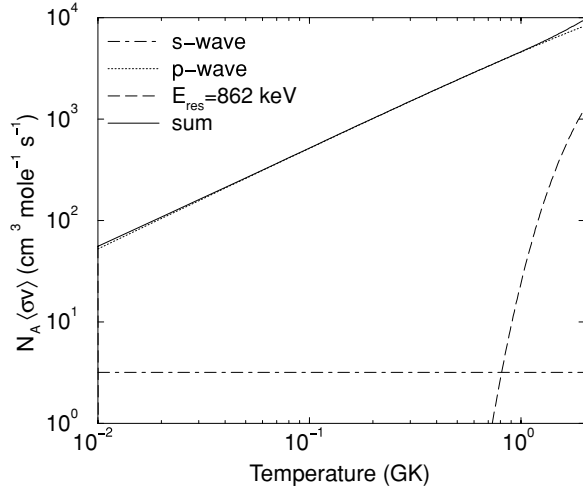


FIG. 5. $^{15}\text{N}(n, \gamma)^{16}\text{N}$ rate as a function of stellar temperature in GK. The rate is dominated by p -wave direct capture.

factors listed in the last two columns of Table I. The reaction rate is dominated by p -wave direct capture to low-lying ^{16}N levels. The calculated p -wave contribution depends directly on the measured spectroscopic factors and thus it was scaled from Ref. [7]. The s -wave contribution is strongly-inhibited since it requires a $E2/M1$ transition; an estimate of its strength from Ref. [20] is also displayed. The only relevant resonance contribution arises from the 862-keV resonance, and its contribution is taken unchanged from Ref. [7]. The resulting reaction rate is plotted in Fig. 5 and is given by the

expression

$$N_A\langle\sigma v\rangle = 3.18 + 5.29 \times 10^3 T - 715 T^{1.7} + 5.3 \times 10^5 T^{-3/2} \exp(-10.0/T), \quad (1)$$

where the rate is given in $\text{cm}^3 \text{mole}^{-1} \text{s}^{-1}$ and the temperature is given in GK. The calculated $^{15}\text{N}(n, \gamma)^{16}\text{N}$ reaction rate is about a factor of 2 larger than the rate from Ref. [7] owing to the larger spectroscopic factors used in the current calculation.

In conclusion, there has been a long-standing discrepancy between the measured neutron spectroscopic factors for ^{16}N and the expected values [7–9]. The spectroscopic factors have a direct impact on the calculated $^{15}\text{N}(n, \gamma)^{16}\text{N}$ reaction rate of importance to nucleosynthesis calculations in AGB stars. We have performed the first re-measurement of the $^{15}\text{N}(d, p)^{16}\text{N}$ reaction, using inverse kinematics with the SIDAR and ORRUBA arrays in conjunction with the Daresbury Recoil Separator. We find that the neutron spectroscopic factors are actually consistent with the near-unity expectations and that the $^{15}\text{N}(n, \gamma)^{16}\text{N}$ reaction rate is roughly a factor of 2 larger than calculated in Ref. [7] as a direct result of these increased spectroscopic factors.

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- [1] S. Wilmes, V. Wilmes, G. Staudt, P. Mohr, and J. W. Hammer, *Phys. Rev. C* **66**, 065802 (2002).
 - [2] M. Lugaro *et al.*, *Astrophys. J.* **615**, 934 (2004).
 - [3] K. Werner, T. Rauch, and J. W. Kruk, *Astron. Astrophys.* **433**, 641 (2005).
 - [4] F. Herwig, N. Langer, and M. Lugaro, *Astrophys. J.* **593**, 1056 (2003).
 - [5] H. T. Fortune and A. G. Lacaze, *Phys. Rev. C* **67**, 064305 (2003).
 - [6] F. de Oliveira *et al.*, *Phys. Rev. C* **55**, 3149 (1997).
 - [7] J. Meissner, H. Schatz, H. Herndl, M. Wiescher, H. Beer, and F. Käppeler, *Phys. Rev. C* **53**, 977 (1996).
 - [8] W. Böhne, J. Bommer, H. Fuchs, K. Grabisch, H. Kluge, and G. Röschert, *Nucl. Phys.* **A196**, 41 (1972).
 - [9] D. W. Lee, K. Peräjärvi, J. Powell, J. P. O’Neil, D. M. Moltz, V. Z. Goldberg, and J. Cerny, *Phys. Rev. C* **76**, 024314 (2007).
 - [10] R. L. Kozub *et al.*, *Phys. Rev. C* **71**, 032801(R) (2005).
 - [11] J. S. Thomas *et al.*, *Phys. Rev. C* **71**, 021302(R) (2005).
 - [12] K. L. Jones *et al.*, *Acta Phys. Pol. B* **38**, 1205 (2007).
 - [13] B. H. Moazen, R. L. Kozub, C. D. Nesaraja, M. S. Smith, J. C. Blackmon, D. W. Bardayan, and J. S. Thomas, *Bull. Am. Phys. Soc.* **48**, No. 8, 56 (2003).
 - [14] J. A. Howard, R. L. Kozub, J. S. Thomas, M. S. Smith, J. C. Blackmon, D. W. Bardayan, and C. D. Nesaraja, *Bull. Am. Phys. Soc.* **49**, No. 6, 57 (2004).
 - [15] D. W. Bardayan *et al.*, *Phys. Rev. Lett.* **83**, 45 (1999).
 - [16] A. N. James *et al.*, *Nucl. Instrum. Methods Phys. Res. A* **267**, 144 (1988).
 - [17] S. D. Pain *et al.*, *Nucl. Instrum. Methods Phys. Res. B* **261**, 1122 (2007).
 - [18] C. M. Perey and F. G. Perey, *At. Data Nucl. Data Tables* **17**, 1 (1976).
 - [19] University of Surrey modified version of the code TWOFNR of M. Igarashi, M. Toyama, and N. Kishida (private communication).
 - [20] W. A. Fowler, G. R. Caughlan, and B. A. Zimmerman, *Annu. Rev. Astron. Astrophys.* **5**, 525 (1967).